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Development of a Mobile-Based Sports Field Booking Application

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Abstract

In the rapidly advancing digital age, mobile-based sports field booking applications offer an innovative solution to the challenges faced by individuals who want to maintain a healthy lifestyle but have difficulty finding and contacting sports venues, which often leads to a decline in their interest in exercising. This sports field booking application was developed to address these issues. The research aimed to investigate the impact of such applications on the accessibility and participation in sports activities. The methodology involved interviewing 20 respondents, including managers and users of sports facilities. The analysis revealed that using this application significantly improves efficiency in schedule management and provides users with greater control over their time. The ease of access offered by the app also increases participation in sports activities, encouraging individuals to stay consistently active. Additionally, the research showed that employing the agile software development lifecycle method in the application's development resulted in a responsive, continuous feedback loop and efficient products. The findings suggest that the sports field booking application has substantial potential to help people exercise regularly and effectively overcome the challenges of booking sports facilities. Future development recommendations include enhancing service quality, integrating a more flexible payment system, expanding the platform, adding a reminder notification feature, and continuously gathering user feedback to improve the overall user experience.

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1. Introduction

Sport has an important role in human life, not only for maintaining physical health, but also for mental and social well-being [1]. In a society that continues to develop, interest in sports activities is increasing. This activity is not only a means of maintaining physical fitness, but also a place for socialization, competition, and an important source of entertainment.

Management of sports facilities and the organization of sports activities are very important. Use of facilities such as soccer, tennis, or basketball fields often requires an efficient booking process [2]. Several studies support the importance of integrating technology in sports facility management. One of the studies states that mobile applications significantly enhance user experience and facility utilization by providing real-time information and streamlined booking processes [3]. A mobile-based sports field booking application offers a relevant and innovative solution by simplifying the booking process, increasing accessibility, and optimizing facility use, allowing people to engage in sports activities more conveniently.

This application is developed using the Agile SDLC method and built with the Dart programming language and Google's Flutter framework [4]. Driven by the demand for efficient access to sports facilities, this research aims to provide solutions to booking challenges. It seeks to promote an active lifestyle and improve the physical, social, and mental health of society.

2. Related Work

2.1. Application System

An application system is a framework consisting of various coordinated elements that aim to achieve certain goals through the use of software [5]. According to previous research, effective application architecture not only enhances performance but also ensures scalability and adaptability to changing user requirements and technological advancements [6]. The application itself is a derivative of software that is designed to perform certain tasks according to user needs [7]. Application architecture, as the overall design and structure, has an important role in ensuring the performance and sustainability of the application [8]. The application life cycle involves stages from planning, development, test, implementation, maintenance, to upgrade [9].

2.2. Android Application Development

The Android operating system, developed by *Google*, is designed specifically for mobile devices such as smartphones and tablets. Based on the Linux kernel, Android has a theoretical foundation that includes several main aspects [10]. One of them is the Android architecture which includes core operating system components such as activity, service, content provider, and broadcast receiver [11]. Based on previous research, which states that the modular design of the Android architecture allows for efficient resource management and flexibility in application development [12].

2.3. Dart

Dart is a programming language developed by Google, designed by Lars Bak and Kasper Lund [13], with the main aim of providing efficient, productive, and easy-to-learn tools for developers [14]. With its flexibility, Dart can be used in various application development contexts, from web applications to server development. According to [15], Dart was developed by Google to overcome the limitations of JavaScript and other web scripting languages, with the goal of being efficient, fast, and secure [16]. One of the studies note that Dart's performance and speed make it particularly suitable for real-time applications and dynamic user interfaces [17].

3. Methodology

3.1. Type of Research

In this research, data collection uses qualitative methods, according to [18] using a qualitative approach can explore aspects that may be difficult to measure quantitatively, such as perceptions, attitudes and experiences. [19] Qualitative research involves collecting and analyzing non-numerical data to understand social phenomena in depth.

3.2. Data Retrieval

In this research, the interview method was used by asking questions to obtain information. According to Fadhallah, RA (2021) Interviews involve conversations between individuals for asking and answering questions. Widely used in job recruitment, journalism, and research, interviews aim to gather information, assess skills, and obtain input. Through interviews, researchers can gather valuable information and input from potential users to develop the desired sports field booking application [20].

3.3. Agile

Agile is a software development methodology that refers to generally accepted and practical principles that can be changed according to the needs of the project and team [21]. Agile emphasizes software development that is centered on individuals and interactions, thus that researchers can work together with high efficiency and make informed decisions [22].



Fig. 1. Stages of Agile Development

In the implementation of this application development, Agile is used as the software development lifecycle method. Planning involves identifying requirements like field search, booking, and payment. Implementation focuses on developing features using Dart. Software testing ensures bug-free functionality. Documentation covers user guides, system architecture, and troubleshooting. Deployment prepares the app for Android. Application performance maintenance involves monitoring, bug fixes, and updates. Figure 1 shows stages of Agile SDLC starting from the requirement phase.

4. Discussion and Result

4.1. Requirement of Interview Results

Interviews were conducted with 20 respondents, encompassing 10 sports facility managers and 10 users who are office workers, to understand their views on the sports field booking application. A user highlighted that the application is very helpful for booking fields efficiently without needing to call or visit in person. The app enhances scheduling control, increases participation in sports activities, and motivates consistent exercise due to its easy access. In an era dominated by technology, sports field booking applications positively impact users by providing easy access and efficient scheduling. Continued technological development is expected to further enhance service quality and user benefits.

4.2. Design

1) Flow Map Diagram

Figure 3 shows the flowmap diagram of the running system in the AnyField application from the start opening the application to the completion of making the sports field order.

2) Use Case Diagram

Figure 3 illustrates the use case diagram explaining the running system process. The user is the person who will use the AnyField application to book a sports court. The manager is the person who uses the AnyField application to receive sports court bookings.

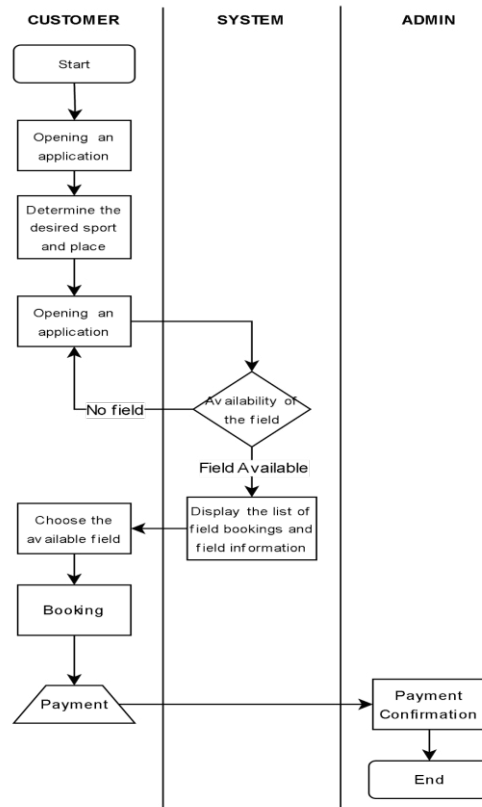


Fig. 2. Flowmap Diagram of Anyfield Application

4.3. Development

In these phases, developing and performing internal application testing were done to ensure its functionality. Dart was used to develop the application, then continue to testing. Testing features such as field booking was to ensure there were no bugs that disrupt the user experience. Following are figures illustrating the results of the AnyField application from Dart development and followed by testing which were passed for all features.

Figure 4 shows the Login page wireframe. This page is the start page when opening the AnyField application. This figure also shows when a user login incorrectly. The application will show a notification when the username or password the user enters is incorrect.

Based on Figure 5, it shows the wireframe for the register page for this application. Users are required to register to be able to access this application. After all registration data has been filled in, you can press the save button at the bottom right.

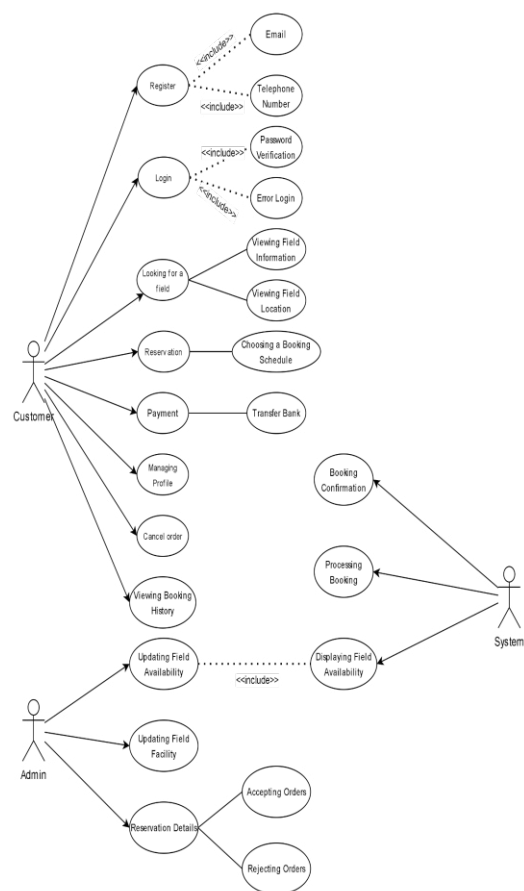


Fig. 3. Use Case Diagram of Anyfield Application

Figure 6 displays a wireframe on the Dashboard page, which contains a list of Badminton, Football, Futsal and Tennis sports fields. Next, users can select the field they want to order.

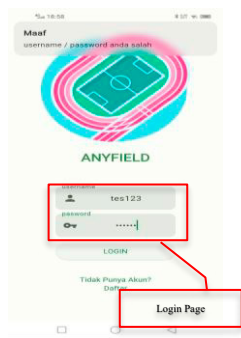


Fig. 4. Login Wireframe

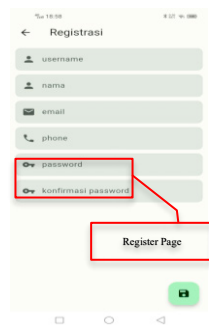


Fig. 5. Wireframe of the Register Page

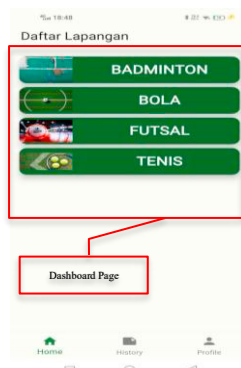


Fig. 6. Dashboard Page Wireframe



Fig. 7. Court Detail Page Wireframe (Badminton)

Figure 7 displays a wireframe on the badminton sports court details page. This page is still connected to figure 6, where when the user selects and clicks on the sport they want, this page will appear and show all types of fields as well as the venue and address.

Figure 8 shows the wireframe of the field rental page. After the user selects the field and address to be booked, the application will immediately show the selected field, the user can immediately select the date, time and duration to be booked.

Figure 9 shows the user page for selecting a booking date. When the user clicks Date, the application will display a list of dates as shown in the image below.

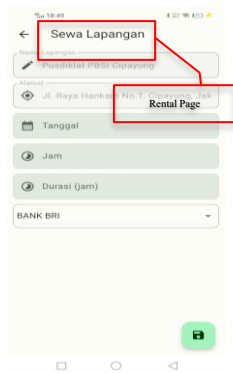


Fig. 8. Wireframe of Field Rental User Page

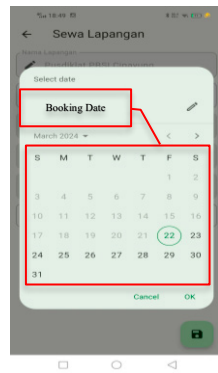


Fig. 9. Wireframe of User Page Selecting Booking Date

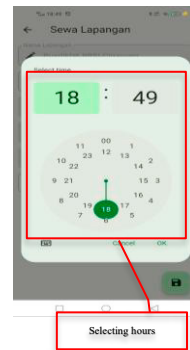


Fig. 10. User Wireframe Selecting Booking Hours

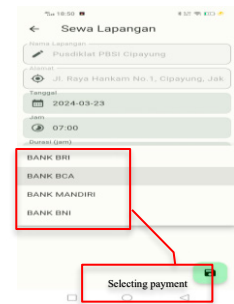


Fig. 11. Wireframe of User Selecting Payment Method

Figure 10 shows the user page for selecting sports field booking times. After selecting the date as shown in Figure 8, the user must select the time to book the sports field they want to reserve.

Figure 11 shows the page for selecting a payment method. After the user selects the date and time, the user can choose a payment method.

Figure 12 displays the wireframe when the user has clicked the save button at the bottom right shown in Figure 11. After that, the user can pay and immediately click rent. However, if the user does not book the field, they can select the cancel button.

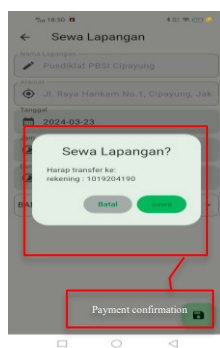


Fig. 12. Wireframe of Buyer Paying and Field Booking

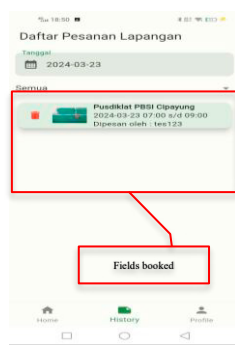


Fig.13. Wireframe Viewing Fields That Have Been Booked

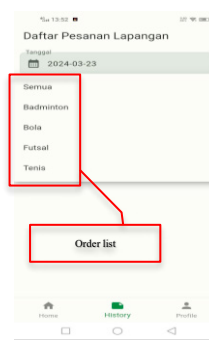


Fig. 14. Field List Wireframe from the Field Order List

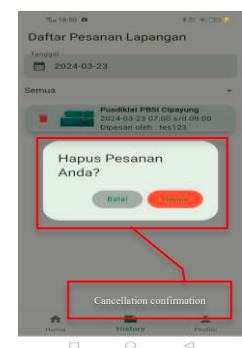


Fig. 15. Admin Wireframe Canceling Field Booking

Figure 13 displays the page when the user sees the field that has been booked. Figure 14 shows a feature for sorting fields which aims to make it easier for users to select fields that have already been booked.

Figure 15 shows the conditions if the admin cancels a user's field booking.

Table 1. UAT Scenario

No	Test Scenario	Test Case	Objective	Steps	Expected Result	Result
1	User Registration and Login	User Registration	Verify that a new user can register successfully	• Navigate to the registration page. • Enter valid user information. • Submit the registration form.	User is registered successfully, and a confirmation email is sent.	passed
2	User Registration and Login	User Login	Verify that registered users can log in	• Navigate to the login page. • Enter valid credentials. • Click the "Login" button	User is logged in successfully and directed to the dashboard	passed
3	Field Search and Availability Check	Search for Available fields	Verify that users can search for available sports fields	• Navigate to the search page. • Enter search criteria (date, location, type of sport). • Click "Search."	A list of available fields matching the criteria is displayed	passed
4	Field Search and Availability Check	Check Field Availability	Verify that users can view the availability of a selected field	• Select a field from the search results. • Check the availability for the desired date and time.	Availability is displayed accurately	passed
5	Field Booking and Payment Process	Book a Field	Verify that users can book a sports field	• Select an available time slot. • Confirm booking details. • Proceed to payment.	Booking is confirmed, and a confirmation message is displayed	passed

4.4. Testing Result

The purpose of UAT is to ensure that the application meets the business requirements and functions as expected in real-world scenarios, with objectives:

- 1) To validate that the application meets the functional requirements.
- 2) To identify and resolve any issues before deployment.
- 3) To confirm that the application is user-friendly and intuitive.
- 4) To ensure the application is ready for production.

The UAT will cover the following key features of the Sports Field Booking Application: User registration and login, Field search and availability check, Field booking and payment process.

All scenarios that is shown in table 1 have passed the UAT and it can continue to the deployment.

4.5. Deployment

Deploy was launched internally within management team and future customers to ensure alpha testing was properly done and it will come to the google store for public usage and might have continue to beta testing.

4.6. Review & Maintenance

Review covers the feedback that received from the end user in deployment phase and maintenance for monitoring the performance of the application and updating new features or improvements in order to make this application successful and can meet user needs.

5. Conclusion

The conclusion of this research, using an agile method with 20 respondents, is that the sports field booking application shows potential in facilitating users to organize and access sports fields. The use of agile methods in application development appears promising, contributing to the creation of responsive and efficient products. The

main takeaway is that this app increases efficiency in schedule management and the application is already usable.

However, it is important to note that this research has certain shortcomings, such as not covering long-term user engagement and satisfaction. These aspects should be considered in future research and development. Suggestions for further development include improving service quality, integration with a more flexible payment system, expanding the platform, adding a reminder notification feature, and continuing to collect feedback from users to improve the overall user experience. Thus, this sports field booking application has great potential to continue to develop and provide greater benefits to the user community.

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